

Hey! This is: $\vee$. $\iff$. Every symbol links to its entry in the legend at the end of this document.

$$a \vee b \stackrel{\text{def}}{\iff} c = 3 \stackrel{\text{def}}{=} 4$$

$$a \vee b \iff c$$

Lattice: A lattice $(X, \sqsubseteq, \sqcup, \sqcap)$ is a poset (partially ordered set), such that $\forall a, b \in X : a \sqcup b$ and $a \sqcap b$ exist (Mine).

Complete Lattice: A complete lattice $(X, \sqsubseteq, \sqcup, \sqcap, \perp, \top)$ is a lattice, such that

1. $\forall S \subseteq X : \bigsqcup S$ and $\bigsqcap S$ exist.
2. X has a least element $\perp$ and a greatest element $\top$.

For any set X , the powerset $(\mathcal{P}(X), \subseteq, \cup, \cap, \emptyset, X)$ is a complete lattice.

Lets write a fancy $\sqsubseteq_b$ or $R \sqcap_b X$, $\sqcap_b$ or $\sqcup_{\#}$!

1 Concrete and abstract

Single objects are marked with `\abstract` and `\concrete`: $X^{\#} \sqsubseteq Y$. Whole expressions switch worlds with `\abstractly` and `\concretely`: $(X, \sqsubseteq, \sqcup, \sqcap, \perp, \top)$ becomes $(X, \sqsubseteq^{\#}, \sqcup^{\#}, \sqcap^{\#}, \perp^{\#}, \top^{\#})$, and $\bigsqcup_{i \in \mathbb{N}} X_i \sqsubseteq^{\#}$ $\nabla^{\#}$ keeps the mark on the operators that have one.

A Galois connection $\mathcal{P}(\mathbb{Z}) \xrightleftharpoons[\gamma]{\alpha} \mathbb{I}$ and a Galois insertion $\mathcal{P}(\mathbb{Z}) \xleftarrow[\gamma]{\alpha} \mathbb{I}$ relate the two worlds via α and γ , with $\alpha \circ \gamma \sqsubseteq \text{id}$.

2 Semantics

$\mathbb{E}[x + 1] \rho = \rho(x) \oplus 1$ for an environment $\rho \in \mathcal{E} = \mathbb{V} \rightarrow \mathbb{I}$, and $\mathbb{S}^{\#}[s] \circ \alpha \sqsubseteq \alpha \circ \mathbb{S}[s]$.

The fixpoint of a statement is $\text{lfp}_{\perp} f = \bigsqcup_{i \in \mathbb{N}} f^i(\perp)$, its abstract counterpart is $\text{lfp}_{\perp^{\#}} f$ and it is reached with a widening $a \nabla b$ and a narrowing $a \triangle b$.

3 Sets

$$\{x \in \mathbb{N} \mid x \leq 3\} \cup \{4\}, |X|, \langle a, b \rangle, [a, b], f|_S, \rho[x \mapsto v].$$

4 Lambda calculus

$(\lambda x. M) N \rightarrow_{\beta} M[x := N]$ and $\lambda x. M x \rightarrow_{\eta} M$ if $x \notin \text{FV}(M)$. The fixpoint combinator unfolds as $\mathbf{Y} f \rightarrow_{\beta} f(\mathbf{Y} f)$, and $\mathbf{SKK} =_{\beta} \mathbf{I}$. Church numerals: $\bar{3} =_{\alpha} \lambda f x. f(f(f x))$.

Typing: $\Gamma, x : \sigma \vdash M : \tau \implies \Gamma \vdash \lambda x. M : \sigma \rightarrow \tau$, polymorphism: $\Lambda X. \lambda x : X. x : \forall X. X \rightarrow X$, big steps: $\cdot \vdash M \Downarrow v$, and evaluation contexts: $C[\cdot]$.

5 Outside of absint notation

Symbols whose name is still free also work in plain math mode: $a \sqcup b \sqsubseteq \top$, or with `\absintuse`: ∇ .

6 Notation

$\vee$ `\lor` — disjunction (also `\vee`)

$\implies$ `\implies` — implication

$\iff$ `\iff` — bi-implication

$\forall$ \forallforall — universal quantifier
 $=$ \eq — equality
 $\emptyset$ \emptysetset — the empty set
 $\in$ \in — membership
 $\notin$ \notin — non-membership
 $\subseteq$ \subseteq — subset
 $\cup, \cup^\#$ \cup — union (also \union)
 $\cap, \cap^\#$ \cap — intersection (also \intersect)
 $\mathcal{P}, \mathcal{P}^\#$ \powerset — powerset (also \P)
 $\sqsubseteq, \sqsubseteq^\#$ \sqsubseq — partial order (also \partof)
 $\leq$ \leq — less than or equal (also \le)
 $\sqcup, \sqcup^\#$ \sqcup — least upper bound (join) (also \lub, \join)
 $\sqcap, \sqcap^\#$ \sqcap — greatest lower bound (meet) (also \glb, \meet)
 $\bigsqcup, \bigsqcup^\#$ \bigsqcup — least upper bound of a family (also \Lub, \Join)
 $\bigsqcap, \bigsqcap^\#$ \bigsqcap — greatest lower bound of a family (also \Glb, \Meet)
 $\top, \top^\#$ \top — greatest element
 $\perp, \perp^\#$ \bot — least element
 $\nabla, \nabla^\#$ \widen — widening
 $\triangle, \triangle^\#$ \narrow — narrowing
 $\rightarrow$ \to — function space
 $\mapsto$ \mapsto — maps to
 $\circ$ \compose — function composition
 id \id — the identity function
 lfp \lfp — least fixpoint
 α \abstraction — abstraction function (also \abstr)
 γ \concretization — concretization function (also \concr)
 $\begin{smallmatrix} \alpha \\ \rightrightarrows \\ \gamma \end{smallmatrix}$ \galois — Galois connection
 $\begin{smallmatrix} \alpha \\ \rightrightarrows \\ \gamma \end{smallmatrix}$ \galoisinsertion — Galois insertion
 $\mathbb{N}$ \N — the natural numbers
 $\mathbb{Z}$ \Z — the integers
 $\mathbb{V}, \mathbb{V}^\#$ \Variables — the variables
 $\mathbb{I}, \mathbb{I}^\#$ \NumericDomain — a numeric domain
 $\mathcal{E}, \mathcal{E}^\#$ \Environments — the environments
 $\rho, \rho^\#$ \Environment — an environment
 $\oplus, \oplus^\#$ \intadd — addition in a domain

$\mathbb{E}, \mathbb{E}^\#$ \RawEvaluateExpr — expression semantics
 $\mathbb{S}, \mathbb{S}^\#$ \RawEvaluateStatement — statement semantics
 λ \lambdalambda — abstraction
 Λ \tylam — type abstraction
 $\backslash\text{app}$ — application
 $\longrightarrow_\beta$ \betared — beta reduction
 $\rightarrow_\beta$ \betareds — beta reduction in many steps
 $=_\beta$ \betaetaeq — beta equality
 $\longrightarrow_\eta$ \etared — eta reduction
 $=_\alpha$ \alphaetaeq — alpha equality
 $\Downarrow$ \evalto — evaluates to
 $\vdash$ \types — entails (also \entails)
 $:$ \hastype — has type
 $\Gamma, \Gamma^\#$ \Context — a typing context
 $\cdot$ \emptyctx — the empty context
 $[]$ \hole — the hole of a context
 I \combI — the I combinator
 K \combK — the K combinator
 S \combS — the S combinator
 Y \combY — the Y combinator